Project Report: Census Income Classification & Customer Segmentation

1. Executive Summary

This project addresses two business objectives:

1. **Classification:** Predict whether an individual earns more or less than $50,000/year.
2. **Segmentation:** Identify distinct customer groups in the population to support differentiated marketing strategies.

The final classifier LightGBM achieves a ROC-AUC of **0.9404** on held-out test data. Four customer segments were identified, each with a distinct profile and marketing implication. The key business recommendation is to use both models in combination: the classifier scores individuals, and the segmentation model guides the messaging strategy per group.

2. Data Exploration

2.1 Dataset Overview

- **Records:** 199,523 observations
- **Features:** 40 demographic and employment variables including age, education, occupation, marital status, capital gains/losses, etc.
- **Weight column:** Each record carries a sampling weight representing how many people in the general population that record represents.
- **Label:** Binary — income ≤ $50K (`- 50000.`) or > $50K (`50000+.`)

2.2 Class Imbalance

Class	Raw Count	Weighted % of Population
≤ $50K	187,141	~93.8%
> $50K	12,382	~6.2%

The dataset is highly imbalanced — roughly 15:1, as shown in Figure 1. This is realistic because high earners are a minority, but requires special handling in model training to avoid the model defaulting to always predicting the majority class.

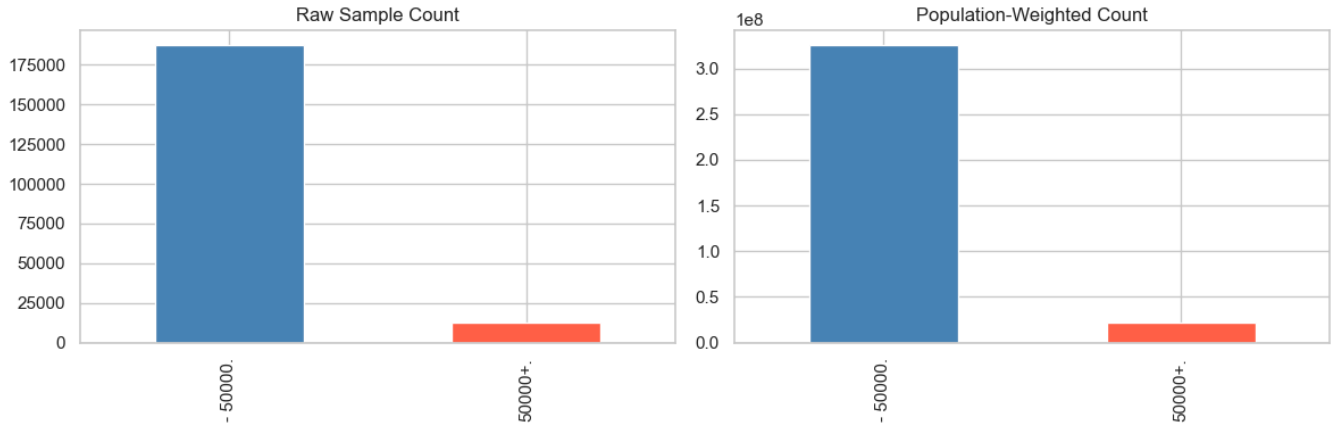

Figure 1: Label distribution showing the 15:1 class imbalance between ≤$50K and >$50K earners.

2.3 Key Findings from EDA

- Education is a strong predictor: college graduates and above have substantially higher rates of earning >$50K (Figure 2).
- Age peaks around 40–55 for >$50K earners; very young and very old individuals rarely exceed the threshold (Figure 3).
- Capital gains and dividends are highly skewed — most values are zero, but when non-zero they strongly indicate high income.
- Missing values appear in 9 columns (represented as ?). Four migration columns have ~99,700 missing values each (~50% of records); the remaining 5 columns including country-of-birth fields, Hispanic origin, state of previous residence have far fewer gaps. All were imputed with mode rather than dropped, for two reasons: dropping would lose up to half the dataset, and the missingness is semantically meaningful — "Not in universe" is the dominant category, so mode imputation correctly assigns these records to the most common valid category rather than inventing a signal.
- Weight-adjusted analysis confirms that the raw sample proportions broadly reflect population distributions, validating use of the full dataset without re-weighting for model training.

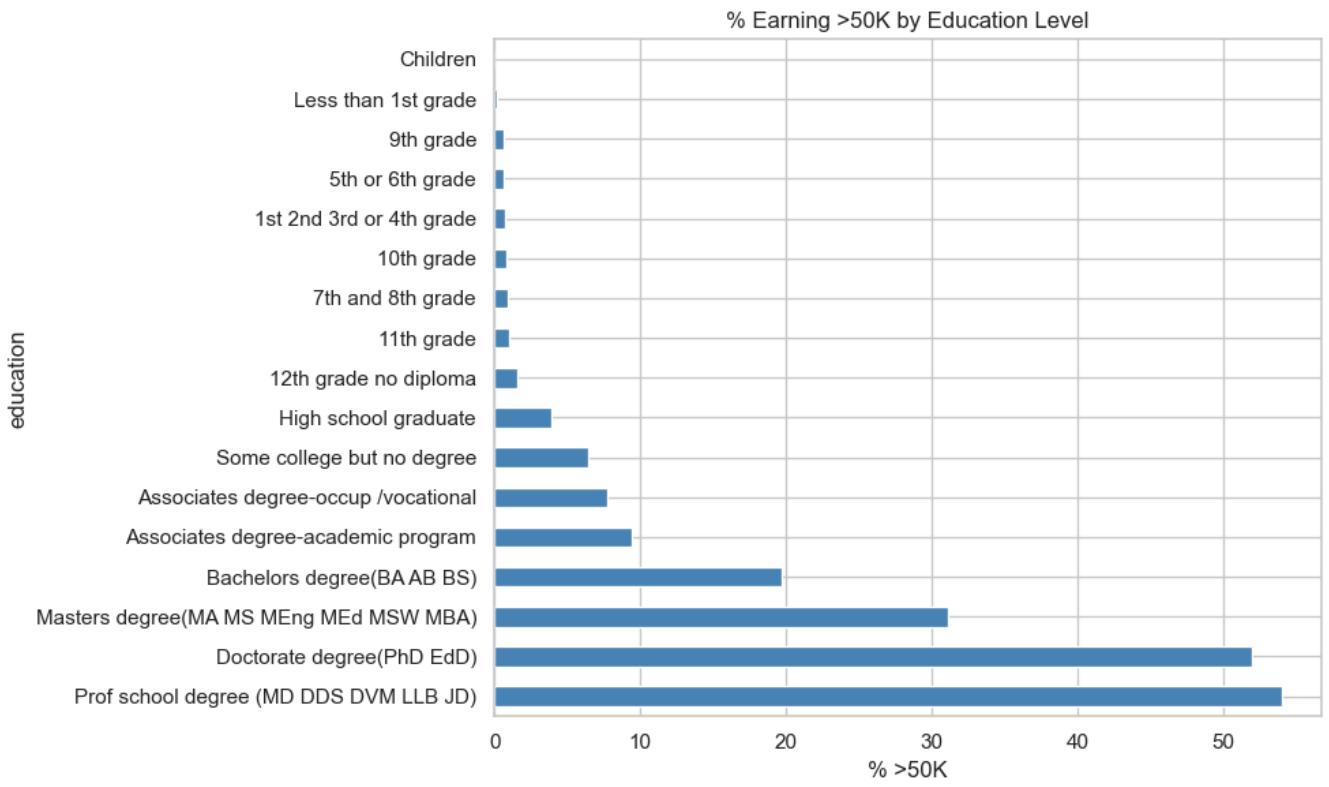

Figure 2: Percentage of individuals earning >$50K by education level.

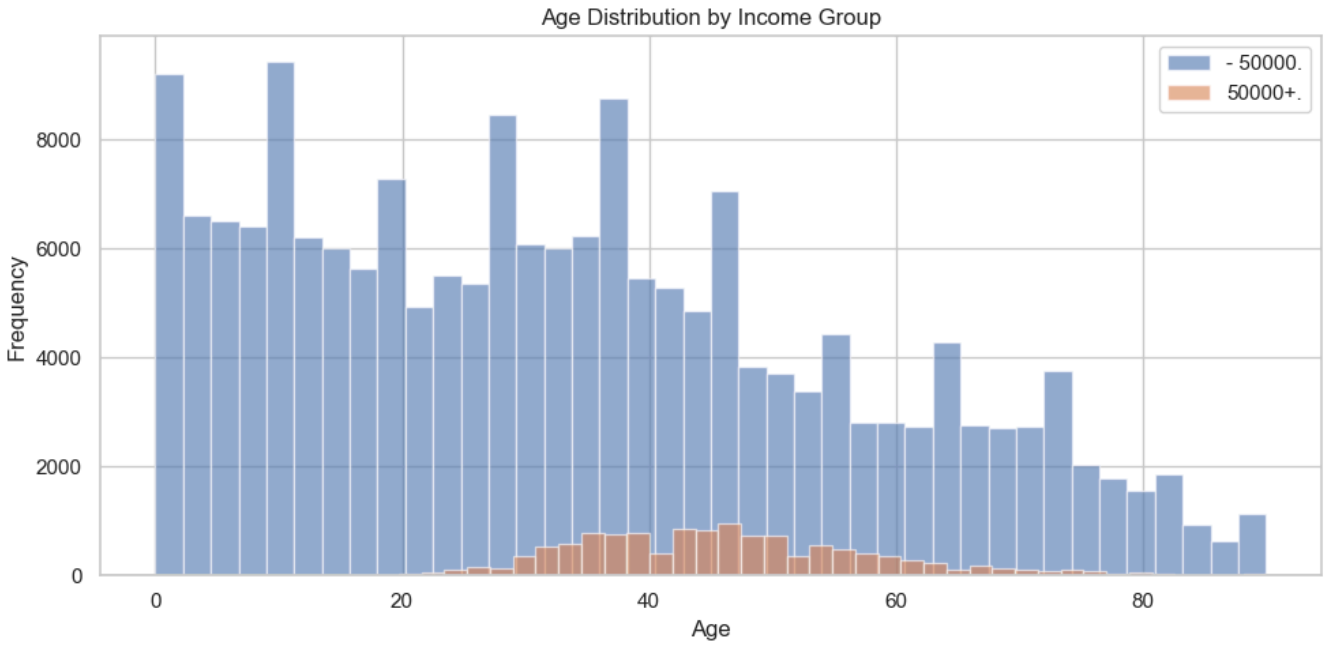

Figure 3: Age distribution by income group.

3. Data Preprocessing

3.1 Decisions Made

Step	Decision	Rationale
Missing values	Impute with median / mode	Retains all 199K rows; missingness is not random but correlates with "Not in universe" categories

Step	Decision	Rationale
Categorical encoding	Label encoding	Compatible with tree models (LightGBM); avoids high-dimensional one-hot encoding for 30+ categorical columns
Weight column	Excluded from features; retained for population-level analysis	Weight is a sampling artifact, not a demographic signal
Label encoding	`50000+.` → 1, `- 50000.` → 0	Standard binary target

3.2 Note on Encoding Choice

I deliberately used label encoding for tree-based models. LightGBM handles label-encoded categorical variables through split points, which avoids problems arising from any implied ordinal structure. For Logistic Regression, we applied StandardScaler to account for the scale differences introduced by this encoding.

4. Classification Model

4.1 Approach

Two models were trained:

- **Logistic Regression** (baseline) — interpretable, establishes a performance floor
- **LightGBM** (main model) — gradient boosting, handles mixed feature types and imbalanced data well

Both use `class_weight='balanced'` / `scale_pos_weight` to address the 15:1 class imbalance.

4.2 Train and Test Split

The data was divided into three stratified subsets: 72% for training, 8% for validation, and 20% for testing. The validation set was used solely for LightGBM early stopping and for selecting the optimal number of trees, while the test set was reserved entirely for final evaluation. The positive rate of 6.2% was maintained across all three subsets.

4.3 Results

Model	ROC-AUC	Avg. Precision
Logistic Regression	0.9303	0.5414
LightGBM	**0.9404**	**0.5722**

Both models achieve strong discriminative power (ROC-AUC > 0.93), as shown in Figure 4. LightGBM is preferred as the production model.

With the default decision threshold of 0.5, LightGBM predicts all observations as ≤50K. This pattern is likely driven by the use of `scale_pos_weight = 15`, which improves the model's attention to the minority class during training but also shifts predicted scores downward. As a result, ROC-AUC remains informative because it reflects ranking performance rather than threshold-based classification, whereas the raw scores should not be interpreted as calibrated probabilities.

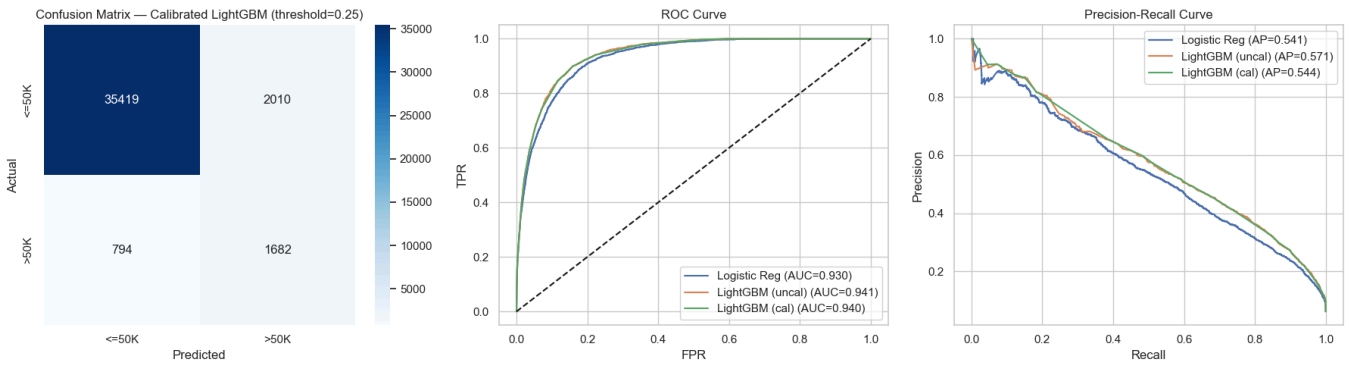

Figure 4: ROC curve and Precision-Recall curve for Logistic Regression, uncalibrated LightGBM, and calibrated LightGBM.

4.4 Probability Calibration

To fix the compressed score distribution, isotonic regression calibration was applied via `CalibratedClassifierCV(cv='prefit', method='isotonic')` using the held-out validation set. The calibrator remaps raw scores to empirical probabilities without retraining the underlying model.

Effect on scores:

	Min score	Max score	Mean score
Uncalibrated	~0.000	~0.320	low
Calibrated	0.0	1.0	~0.062

After calibration, the reliability diagram (Figure 5) shows the model is well-calibrated — predicted probabilities track the actual fraction of positives across all probability bins. ROC-AUC is preserved since calibration does not change ranking order.

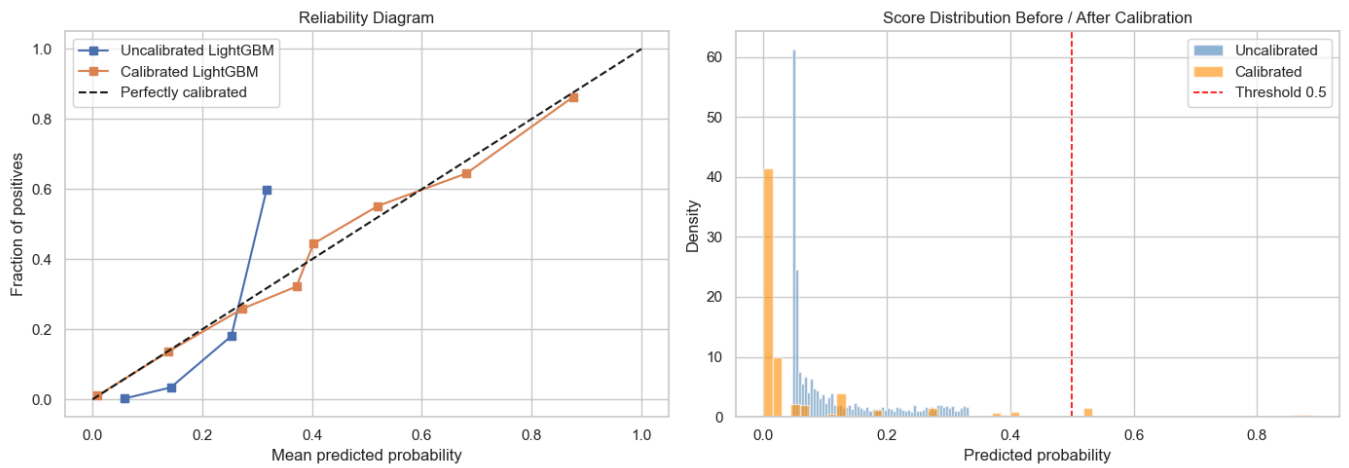

Figure 5: Left — reliability diagram comparing uncalibrated and calibrated LightGBM. Right — score distribution before and after calibration.

4.5 Threshold Selection for Marketing

All thresholds below refer to **calibrated probabilities**, which are now directly interpretable as "probability this person earns >$50K."

Threshold	Precision (>50K)	Recall (>50K)	# Flagged (test set)
0.10	16.5%	97.3%	14,609
0.15	22.6%	93.5%	10,260
0.20	28.9%	87.5%	7,508
0.25	39.1%	77.3%	4,898
0.30	60.1%	48.0%	1,980

interpretation:

- Broad campaign (threshold 0.20): Flag ~7,500 people per 40K; 87.5% of true high-earners captured, ~29% of flagged are genuinely high-income.
- Precision targeting (threshold 0.30): Flag ~2,000 people; 60% of flagged are genuinely high-income, but miss ~52% of the high-earner pool.
- Unlike the uncalibrated model, which collapses to zero flagged above 0.35, after calibration, thresholds above 0.30 continue to yield predictions, enabling finer precision targeting if needed.

Recommendation: Use threshold 0.20–0.25 for direct mail / digital campaigns, and threshold 0.30 for high-touch channels (e.g., personal outreach, premium offers) where cost-per-contact is high. Always apply the calibrated model in production so that scores are interpretable as actual probabilities.

4.6 Top Predictive Features

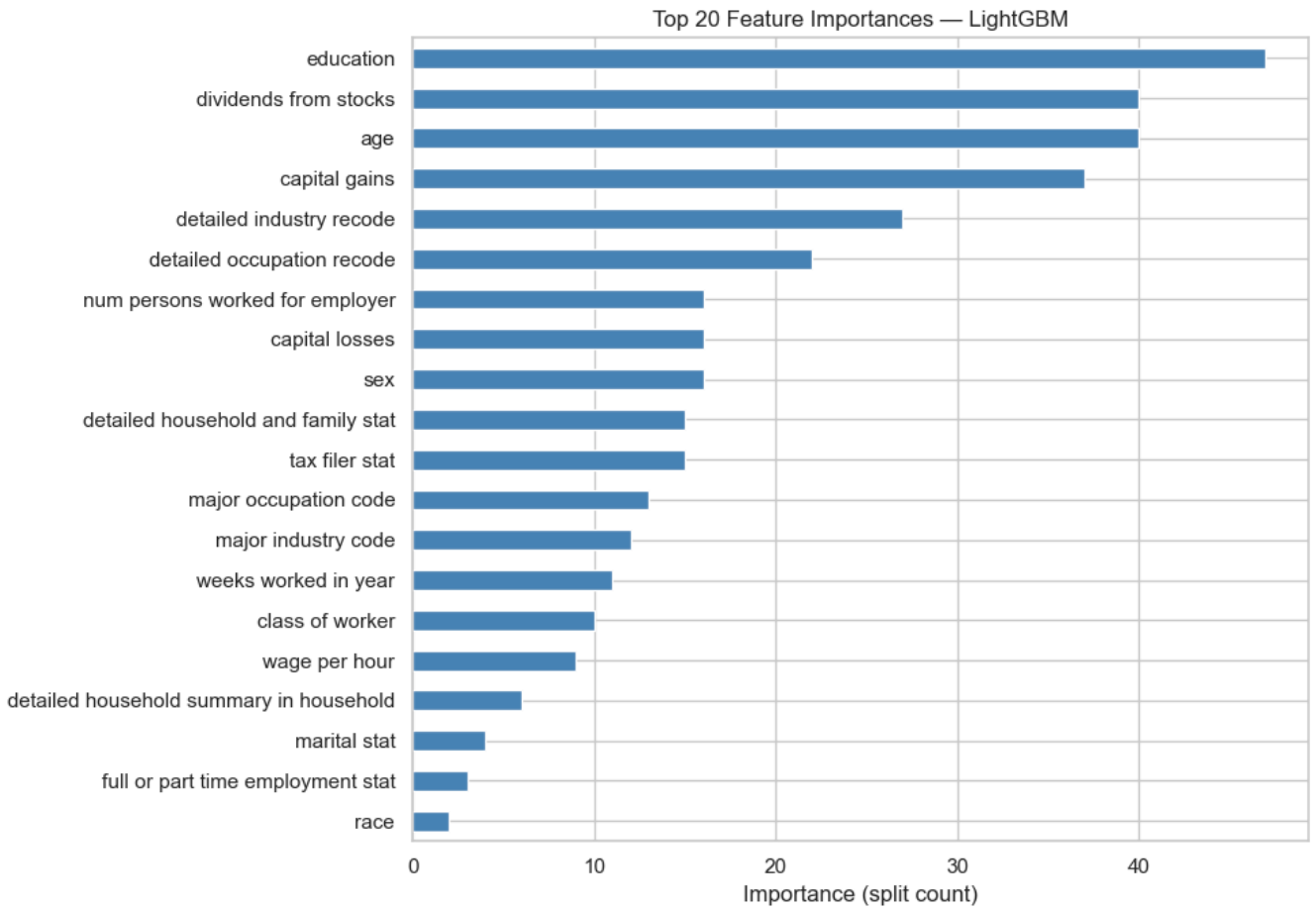

Figure 6: Top 20 features by LightGBM split importance.

Investment income (dividends, capital gains) and human capital (education, occupation) dominate, as shown in Figure 6. A notable insight is that the top two features are both investment income, not salary or occupation. This means the model is primarily identifying *investors* rather than high-salaried employees — a meaningful distinction for the client. People crossing the $50K threshold in this dataset tend to have non-zero capital market exposure, not just high wages. This should inform product strategy: financial and investment products are likely more relevant to this population than purely lifestyle-based premium goods.

5. Segmentation Model

5.1 Approach

Customer segmentation was performed using K-Means clustering on the full feature set excluding label and weight, after dimensionality reduction via PCA.

Why PCA first?

- 40 features with label-encoded categoricals create a mixed, high-dimensional space where Euclidean distance is less meaningful.
- PCA de-correlates features and focuses clustering on the principal axes of variation.
- 20 components were retained, explaining 80.8% of total variance.

5.2 Choosing K

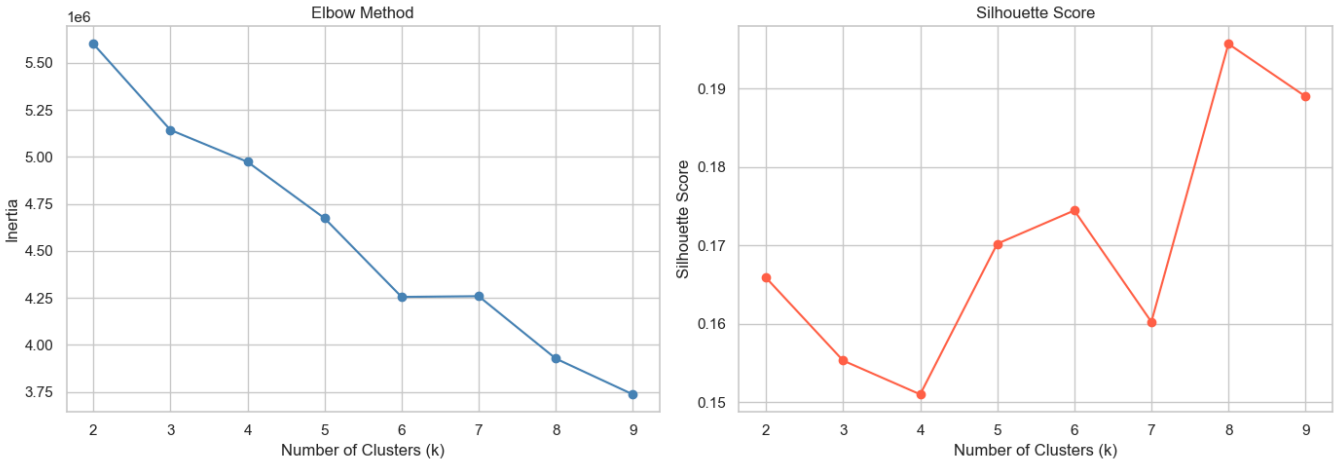

Figure 7: Elbow method (inertia) and silhouette scores for k = 2 to 9.

Elbow method and silhouette scores were evaluated for k = 2 to 9 (Figure 7). Silhouette scores for k = 4 are approximately 0.17, rising to ~0.19 at k = 5–6. K = 4 was selected as a deliberate business decision, not because the statistics are indistinguishable. Inspecting k = 5 and k = 6 clusters showed that the additional segments split existing life-stage groups without producing profiles that map to distinct marketing actions. Four segments — children/non-working, prime working adults, near-retirement adults, and moderate mid-career earners — correspond to meaningfully different communication strategies and are directly actionable for the client.

5.3 Cluster Profiles

Cluster sizes are reported as population-weighted percentages.

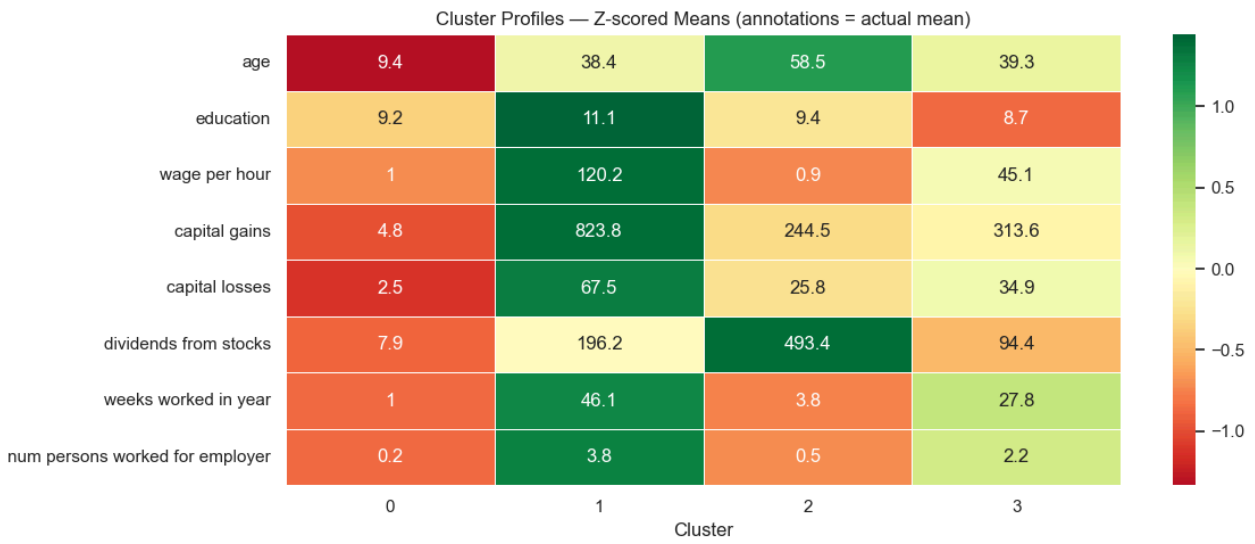

Figure 8: Cluster profiles heatmap — z-scored feature means across the four segments.

Figure 9: Percentage of individuals earning >$50K by cluster.

Cluster	Pop. Share	Avg. Age	Avg. Wage/hr	Avg. Capital Gains	% Earning >$50K
0	28.0%	~9	~$1	~$5	0.0%
1	44.8%	38.4	$120	$824	12.4%
2	20.5%	58.5	~$1	$244	2.4%
3	6.8%	39.3	$45	$314	5.4%

Segment descriptions:

Cluster 0 — Children / Non-Working (28%) Age profile ~9 years. This group represents minors and non-participants in the labor force. Income is essentially zero. Not a direct marketing target for income-based products, but relevant for family-oriented or education product lines.

Cluster 1 — Prime Working Adults (44.8%) The largest and most economically active segment. Average age 38, higher wage per hour, notable capital gains. Highest share of >$50K earners (12.4%), as shown in Figure 9. This is the primary target for premium retail, investment products, and lifestyle brands.

Cluster 2 — Older / Near-Retirement Adults (20.5%) Average age 58.5, lower wages but moderate capital gains (likely from accumulated investments). Income >$50K rate is low (2.4%), but this group may have significant net worth not reflected in current income. Relevant for retirement financial products, health & wellness, and leisure/travel.

Cluster 3 — Mid-Career Moderate Earners (6.8%) Smallest population segment. Age similar to Cluster 1 but lower wages and lower >$50K rate. Likely includes part-time workers, self-employed with variable income, or career-changers. Relevant for affordable mid-market retail and financial education products.

6. Business Recommendations

6.1 Combined Model Strategy

Use both models together:

1. Score everyone with the classifier to get a probability of earning >$50K.
2. Assign everyone to a customer segment.
3. Prioritize Cluster 1 individuals with classifier score ≥ 0.25 for premium outreach — at this threshold, ~39% of flagged individuals are genuine >$50K earners and ~77% of all >$50K earners in the population are captured (see Section 4.5).
4. Customize messaging by segment even for lower-scoring individuals — a Cluster 2 member with a low score may still have significant net worth, warranting a different message than a Cluster 3 member with the same score.

6.2 Segment-Specific Marketing

Segment	Recommended Channel	Product Focus
Cluster 1 (high score)	Premium direct mail, digital retargeting	Luxury goods, investment products, travel
Cluster 1 (low score)	Broad digital campaigns	Mid-range retail, financial savings products
Cluster 2	Email, print	Retirement planning, health, leisure
Cluster 3	Budget digital channels	Promotions, value retail
Cluster 0	Target parents (Cluster 1/2)	Children's products, education

6.3 Model Limitations & Future Work

- **Data age:** The dataset is from 1994–1995. Income thresholds, industry distributions, and demographics have changed significantly. Re-training on recent data is essential before deployment.
- **$50K threshold:** In 1994 dollars, $50K is roughly equivalent to $100K+ in 2024 terms. The business should consider whether this threshold still represents "high income" for its marketing purposes.

- **Label encoding:** For a production system, target encoding or embeddings for high-cardinality categoricals would improve model quality.
- **Segmentation stability:** K-Means is sensitive to initialization. As an additional validation, we re-ran segmentation on the working-age subset (age ≥ 16) using both KMeans and Gaussian Mixture Models (GMM), comparing silhouette scores across k = 2–8 (see Notebook 03, Section 8). Neither approach improved on the original result: working-age KMeans peaked at silhouette ≈ 0.15 (vs. 0.17 for the original), and GMM achieved 0.31 only at k = 2 — a binary split too coarse for marketing use, with scores collapsing at higher k. This confirms that Cluster 0 (children) represents genuine natural separation rather than an artifact inflating the score, and validates the original k = 4 full-population model as the preferred approach.
- **Ethics:** Features like race and sex are present in the data. The model should be audited for disparate impact before any production use.

7. References

1. U.S. Census Bureau. *Current Population Survey, March 1994 and 1995*. Washington, DC: U.S. Department of Commerce.
2. Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems 30*.
3. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *JMLR 12*, pp. 2825–2830.
4. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.